5.11.1 Constructing a trapezium when its three sides and one diagonal are given

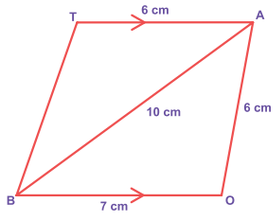
Example 5.26

Construct a trapezium **BOAT** in which $\overline{BO}$ is parallel to $\overline{TA}$, BO=7*cm*, OA=6*cm*, BA=10*cm* and TA=6*cm*. Also find its area.

Solution:

Given:

BO=7*cm*, OA=6*cm*, BA=10*cm* ,
TA=6*cm* and $\overline{BO} \parallel \overline{TA}$



Rough diagram

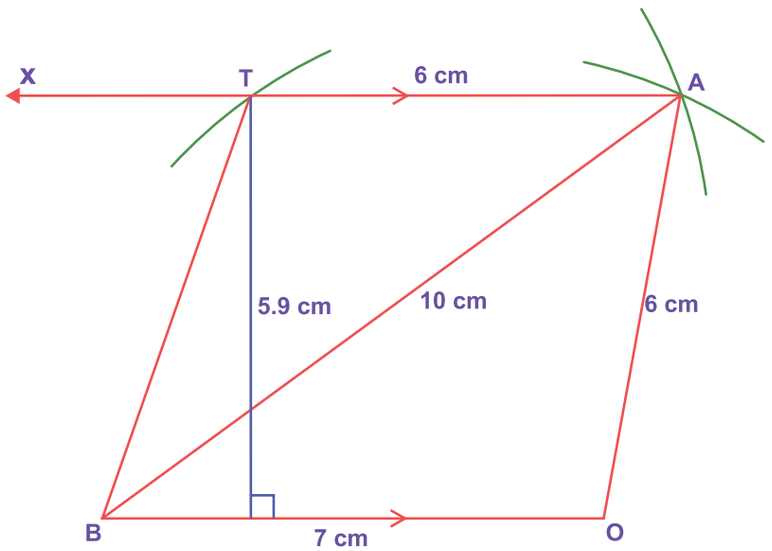


Fig. 5.45

Steps:

1. Draw a line segment BO = 7*cm*.
2. With B and O as centres, draw arcs of radii 10*cm* and 6*cm* respectively and let them cut at A.
3. Join BA and OA.
4. Draw AX parallel to BO
5. With A as centre, draw an arc of radius 6*cm* cutting AX at T.
6. Join BT. BOAT is the required trapezium.

Calculation of Area:

Area of the trapezium BOAT $= \frac{1}{2} \times h \times (a+b)\, sq.units$

$= \frac{1}{2} \times 5.9 \times (7+6) = 38.35\, sq.cm$

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